Report R27 — the evolution operator's Schmidt rank

What the literature establishes, and why a linear-dimensional module forces linear growth

rev. 3: the ceiling, the reflection mechanism, and ranks without a threshold

Tier 1 analysis

2026-08-23

Abstract

Rule 156 has an operator Schmidt rank $\chi(t)$ of its *evolution operator* that grows linearly in t, in a clean, disorder-free, translation-invariant Floquet circuit. This report establishes what the literature says about that observable and supplies a mechanism. §1 is an audited pass over the twelve references supplied plus eleven found by search, sorted by relevance, each with the finding that bears on our claim. **The baseline is Dubail's**: the OSEE of $U(t) = e^{-iHt}$ grows *linearly*, hence χ *exponentially*, "unless the Hamiltonian is in a localized phase". Zhou and Luitz classify the three observed growth laws for $U(t)$ — linear, power law, logarithmic — and both sub-linear classes occur only with disorder. Every disorder-free sub-linear result we could find, including Alba–Dubail–Medenjak for Rule 54 and the very recent Tan–Prosen clean non-integrable circuit, is about *local operators in the Heisenberg picture*, which Dubail explicitly separates from the propagator. We found no disorder-free system with sub-linear evolution-operator OSEE. §2 then derives the mechanism. The key quantity is not integrability but the **number of defects that can cross the cut**: if the sector containing the cut is spanned by configurations of p domain walls with p bounded, then $\chi(t) = O(t^p)$, and if p is extensive χ grows exponentially. The kink module of W_{156}/W_{198} has $p = 1$, giving $\chi(t) = O(t)$; the hard-core module of W_{201}/W_{108} has extensive p, giving exponential growth. Fragmentation into modules of polynomial dimension is therefore a disorder-free mechanism for the localization signature.

New in rev. 2. §3 closes the gap left open in rev. 1. The upper bound is sharpened to an exact combinatorial statement — a *last-passage identity* valid for any operator and any splitting, giving $\mathrm{rank}(P_R A^t P_L) \leq t \cdot \mathrm{rank}(P_R A P_L)$ — and the cross transfer of the kink walk is shown to be *rank one*, in a cell basis in which the walk is a two-band chiral quantum walk. A matching lower bound follows from an explicit rank-t factorisation whose two vector families are independent by causality. The result is the two-sided bound $2t \leq \chi_{\mathrm{module}}(t) \leq 2t+2$, so $\chi = \Theta(t)$, with the measured value exactly $2t+1$ for both rules, every chain length and every cut tested. For the *full* chain the exact law is $\chi(t) = 6t - 7$ for $4 \leq t \leq N/2$, verified at $N = 8, 10, 12, 14, 16$; the slope 6 itself is recorded as measured, together with three ruled-out derivations of it.

New in rev. 3. Three additions, all resting on exact arithmetic. §4: χ does not grow forever, and the ceiling is closed-form — $(N^2+15)/4$ for W_{156}/W_{198}, $F(k)F(k+1)$ for W_{108}, $F(k)\big(F(k)+3F(k-1)\big)$ for W_{201} — confirmed here at $N = 7, 9, 11, 13$ by exact rank over $\mathbb{Q}(\sqrt{2})$ with no cutoff anywhere. *A ceiling polynomial in N and uniform in t is a sharper contradiction of the exponential baseline than any statement about a growth window, and it answers the objection that a linear window is merely a light cone before saturation.* §5: the slope acquires a mechanism — kink reflection between stationary walls — with the measured law slope $= 2(k-1)$ in the number k of straddling-gap length classes, N-independent across $N = 12, 14$. §6: the growth window is narrowed to $4 \leq t \leq \min(N/2, 12)$, and a methodological warning is recorded that cost this project three false results — a *fixed* singular-value cutoff manufactured a feature in three independent pipelines, in both directions.

Contents

1 Protocol: the literature, sorted by relevance

Two objects must be kept apart throughout, because the literature treats them very differently and conflating them is the trap this project already fell into once:

- **(L)** the operator entanglement of a *local* operator $O(t) = U^\dagger O U$ in the Heisenberg picture;
- **(U)** the operator entanglement — equivalently the Schmidt rank of the Choi vector, equivalently the MPO bond dimension — of the *evolution operator* $U(t)$ itself.

Our measurement is **(U)**. Sub-linear results for (L) do not bear on it.

1.1 Tier 1: directly on the claim

reference	object	finding that bears on our claim
Dubail, J. Phys. A **50**, 234001 (2017)	**(U)**	**The decisive baseline.** "The OSEE of the evolution operator $U(t) = e^{-iHt}$ increases linearly with t, unless the Hamiltonian is in a localized phase." Explicitly contrasts this with local operators, whose OSEE "grows sublinearly (perhaps logarithmically)". Linear OSEE $\Rightarrow \chi$ exponential. *Localization is named as the only exception.*
Zhou & Luitz, PRB **95**, 094206 (2017)	**(U)**	Classifies the growth of operator EE *of the time evolution operator* across chaotic, MBL and Floquet systems: linear, power-law and logarithmic growth are all observed, before extensive saturation. Crucially the two sub-linear classes are the *random-field* Heisenberg model — power law at weak disorder, logarithmic at strong disorder. The most chaotic Floquet model saturates at the Page value.
Thakur, Pain & Roy, PRB **111**, 174206 (2025)	**(U)**	Operator entanglement of the time-evolution operator in the MBL phase, showing a *logarithmic light cone*, derived from eigenstate correlations without invoking ℓ-bits. This is the closest published phenomenology to ours — same object, same functional form — but obtained *from disorder*.
Prosen & Žnidarič, PRE **75**, 015202 (2007)	(L)/state	The origin of the exponential-versus-polynomial dichotomy for MPO bond dimension: $D_\epsilon(t) \propto e^{ct}$ in quantum chaotic systems, polynomial when integrable. *But the evolved objects are observables and thermal states, not the propagator*, so this does not license "integrable $\Rightarrow$ polynomial χ" for $U(t)$.

1.2 Tier 2: the disorder-free sub-linear results, all for local operators

reference	object	finding
Alba, Dubail & Medenjak, PRL **122**, 250603 (2019)	(L)	Logarithmic upper bound on operator entanglement for *all local operators* in the Rule 54 chain, and evidence that log growth is generic for interacting integrable systems. A cellular automaton, so tempting — but the wrong object.
Tan & Prosen, arXiv:2603.19363 (2026)	(L)	"Logarithmic growth of operator entanglement in a *clean non-integrable* circuit". Semi-ergodic dual-unitary brickwork; mechanism is that the Heisenberg evolution of a single traceless single-site operator lies in a restricted subspace. **The nearest disorder-free precedent in spirit** — log growth from a restricted subspace rather than from integrability or disorder — but again for a local operator, not $U(t)$.

reference	object	finding
Prosen & Pižorn, PRA **76**, 032316 (2007)	(L)	Transverse Ising chain: OSEE of local operators *saturates* for finite Majorana index and grows *logarithmically* for infinite index. Free fermions; local operators.
Pal & Lakshminarayan, PRB **98**, 174304 (2018)	**(U)**	Entangling power of time-evolution operators, integrable versus non-integrable. The integrable kicked transverse-field Ising chain still shows *ballistic* growth of operator von Neumann entropy; its entangling power saturates lower than the non-integrable case but the growth is not sub-linear. *Integrability alone does not buy sub-linear* $\chi(t)$.

1.3 Tier 3: framework, and the closest methodological neighbour

reference	object	finding
Merbis & Bakker, ICCS 2024, arXiv:2406.04895	**(U)**, classical	**The closest method**: builds the MPO of a cellular automaton's transition matrix and classifies elementary CA by operator-entanglement growth. Types I (converges to a constant, e.g. their rules 136, 108), II (rises then falls, peak at $t \sim L$), III (linear growth to an L-dependent plateau, chaotic-like). Generic bond dimension quoted as 4^t, i.e. $S(t) = 2t$. *No logarithmic class appears.* Their object is the classical transition matrix of a deterministic CA, not a quantum evolution operator with Hadamards, so the rule numbers are not comparable to ours despite the coincidence.
Zanardi, Zalka & Faoro, PRA **62**, 030301 (2000); Zanardi, PRA **63**, 040304 (2001)	**(U)**	Where the object comes from: entanglement of a unitary regarded as a vector in Hilbert–Schmidt space, and the entangling power. Definitional; sets the measure we and everyone else use.
Jonay, Huse & Nahum, arXiv:1803.00089 (2018)	(L)/(U)	The membrane / line-tension coarse-graining. Predicts *linear* growth for generic chaotic dynamics with rate set by the surface tension $\mathcal{E}(v)$. Our result sits outside this picture, which is the point.
Hahn, Luitz & Chalker, PRX **14**, 031029 (2024)	**(U)**	Eigenstate correlations of the time-evolution operator beyond ETH; uses Floquet models without conserved densities to isolate scrambling. Provides the eigenstate-correlation language in which Thakur–Pain–Roy then derive the MBL log light cone.
Bandyopadhyay & Lakshminarayan, arXiv:quant-ph/0504052 (2005)	**(U)**	Entangling power of quantum chaotic evolutions; early evidence that chaotic evolutions approach the random-matrix value.

reference	object	finding
Musz, Kuś & Życzkowski, PRA **87**, 022111 (2013)	**(U)**	Geometry of unitary gates, perfect entanglers, unistochastic maps. Fixes what "maximal" means for the measure.
arXiv:2603.05656 (2026), *Classical simulability from operator entanglement scaling*	(L)	Simulability classification by the scaling exponent α: $\alpha \geq 1$ blocks efficient MPO approximation, $\alpha < 1$ permits it over restricted state ensembles. Local-operator entanglement.

1.4 Verdict of the pass

1. For the *evolution operator*, the established law is linear OSEE, i.e. **exponential** $\chi(t)$, with exactly one documented exception: **localization**, which requires disorder (Dubail; Zhou–Luitz; Thakur–Pain–Roy).
2. Integrability is *not* an exception for this object. Prosen–Žnidarič's polynomial result is for observables and states; Pal–Lakshminarayan find ballistic operator-entropy growth even for an integrable kicked Ising chain.
3. Every disorder-free sub-linear result we found — Alba–Dubail–Medenjak, Prosen–Pižorn, Tan–Prosen — is for *local operators*, which Dubail explicitly separates from the propagator.
4. We found **no** report of sub-linear evolution-operator Schmidt-rank growth in a clean, disorder-free system. A linear $\chi(t)$ for $U(t)$ in a translation-invariant Floquet circuit is, on this evidence, unoccupied ground.

Remark 1. The nearest conceptual precedent is Tan–Prosen: log growth in a clean non-integrable circuit because the evolved operator is confined to a *restricted subspace*. That is the same shape of mechanism as ours — confinement to a small module rather than integrability or disorder — applied to the other object. It should be cited as the closest relative and distinguished on the object.

2 The mechanism: defect counting bounds the Schmidt rank

We now give the argument that a module of linear dimension forces linear $\chi(t)$. The statement turns out not to be about dimension as such, but about *how many defects can cross the cut.*

2.1 Setup

Let U be one period of a brick-wall circuit with strict light-cone velocity v (sites per period). Fix a cut at bond x and let $\chi_x(t)$ be the operator Schmidt rank of $U(t) = U^t$ across it — equivalently the rank of the Choi vector $|U(t)\rangle\rangle$ under the bipartition, equivalently the MPO bond dimension at that bond.

The generic bound is the light cone: $U(t)$ acts as the identity outside a causal diamond of width $2vt$ about the cut, so $\chi_x(t) \leq 4^{2vt}$. Everything in §1 that grows exponentially is saturating a bound of this form. To do better one needs the dynamics to explore only a small part of the operator space on the diamond.

2.2 The kink chain

Take W_{156}. Its sectors are wall-free gaps, and inside a gap the state space is spanned by the *kink states*

$$|k\rangle \;=\; \underbrace{1\cdots1}_{k}\,\underbrace{0\cdots0}_{l-k}, \qquad k=0,\dots,l,$$

of dimension $l+1$ — one degree of freedom, the position of a single domain wall. Restricted to the gap,

$$U(t)\big|_{\text{gap}} \;=\; \sum_{k,k'} A_{kk'}(t)\,|k\rangle\langle k'|,$$

a *single-particle propagator*. Two facts make the Schmidt rank collapse.

Every kink state is a product across the cut. Cutting at x,

$$|k\rangle = |L_k\rangle\otimes|R_k\rangle, \qquad \begin{cases} L_k = 1^k0^{x-k},\; R_k = 0^{\,l-x} & k\le x,\\ L_k = 1^x,\; R_k = 1^{\,k-x}0^{\,l-k} & k>x. \end{cases}$$

So each term $|k\rangle\langle k'|$ is already a product operator $\ell_{kk'}\otimes r_{kk'}$ with $\ell_{kk'} = |L_k\rangle\langle L_{k'}|$ and $r_{kk'} = |R_k\rangle\langle R_{k'}|$. No entanglement is generated by the basis itself.

Three of the four index sectors are rank one. Split the sum by the side of the cut:

- $k,k'\le x$: then $R_k = R_{k'} = 0^{\,l-x}$, so *every* term shares the same right factor. All of these together contribute Schmidt rank ≤ 1.
- $k,k'>x$: then $L_k = L_{k'} = 1^x$, the same left factor. Rank ≤ 1.
- $k\le x<k'$ and $k'\le x<k$: the crossing sectors. Here the left factor is $|L_k\rangle\langle 1^x|$, labelled by k alone, and the right factor is $|0^{\,l-x}\rangle\langle R_{k'}|$, labelled by k' alone. The contribution is the rank of the coefficient block $A_{kk'}$ restricted to that range.

Causality restricts the crossing blocks: $A_{kk'}(t)=0$ unless $|k-k'|\le 2vt$, and a term contributes across the cut only if the kink is within the light cone of x, i.e. $|k-x|,|k'-x|\lesssim vt$. Each crossing block is therefore at most $(vt)\times(vt)$, of rank at most vt. Hence

$$\boxed{\chi_x(t) \;\le\; 2+2\,vt \;=\; O(t).} \tag{1}$$

2.3 The general statement

Nothing above used δ, the Hadamard, or integrability. What was used is that the sector is labelled by the position of *one* defect.

Proposition 1 (Defect counting)**.** *Suppose U preserves a sector whose basis states are labelled by the positions of p defects on the chain, each defect state being a product state across any cut, and suppose U has strict light-cone velocity v. Then the operator Schmidt rank of $U(t)$ across a cut obeys*

$$\chi_x(t) \;=\; O\big((vt)^p\big).$$

If instead the number of defects is extensive in the region size, the bound degrades to the light-cone bound and $\chi_x(t)$ may grow exponentially.

Sketch. As above: a configuration is fixed by the p defect positions, so a basis operator $|c\rangle\langle c'|$ factorises across the cut with left factor determined by the defect positions to the left and right factor by those to the right. Terms in which no defect lies within the light cone of the cut share a common factor on one side and contribute rank 1. Terms with $j\le p$ defects inside the light cone are labelled by at most $(2vt)^j$ positions on each side, so the corresponding coefficient block has rank at most $(2vt)^j$. Summing over $j\le p$ gives $O((vt)^p)$. □

2.4 The two families

rules	gap module	dimension	defects p	predicted $\chi(t)$
W_{156}, W_{198}	kink chain 1^a0^b	$l+1$, linear	1	$O(t)$, **linear**
W_{201}, W_{108}	hard-core chain	$\sim \varphi^l$, exponential	extensive	exponential

This reproduces both measured behaviours: linear $\chi(t)$ for rule 156, exponential for 201/108. It also explains why the split is so sharp despite the two families differing by a single letter of the gate word: the V-context decides the wall word, the wall word decides whether a gap admits an ascent, and that decides whether the gap carries one defect or extensively many.

Remark 2 (What the argument does and does not give). It gives linearity, and the identification of the exponent with the defect number. It does *not* give the measured slope 6. It is an upper bound, so it does not by itself prove the growth is not slower — §3 supplies the matching lower bound. And it assumes the sector basis is product across the cut, which holds for kink and hard-core configurations but must be checked for any other module.

Remark 3 (Why this is the localization signature without localization). In an MBL system the log light cone comes from an emergent set of quasi-local integrals of motion which confine information transport. Here the confinement is kinematic: the sector simply has too few configurations within the light cone for the Schmidt rank to grow. Fragmentation supplies, without disorder, the thing localization supplies with it — a bound on how many distinguishable histories can cross a cut in time t. That is the sense in which Proposition 1 is the mechanism the literature pass says is missing.

3 The tight bound

Proposition 1 gives $\chi = O(t)$. This section replaces it, for the kink module, by a two-sided bound with an exactly matching measurement. All numbers below come from `analytics/kink_schmidt.py`, which rebuilds the propagator from the project's exact $\mathbb{Z}[1/\sqrt{2}]$ branch machinery, not from any earlier tabulation. One bookkeeping point: at obc0 the maximal one-wall Krylov sector of an N-site chain is $\{|k\rangle = 1^k0^{N-k} : k = 1, \ldots, N\}$, of dimension $l = N$ — the all-zero configuration is frozen and sits outside it — so l and N coincide throughout this section. Closure is exact: the amplitude leaking out of the module is 0.0, not merely small.

3.1 Step 1: the reshuffle collapses onto the coefficient matrix

Fix the cut at bond x and recall from §2 that $|k\rangle = |L_a\rangle \otimes |R_b\rangle$ with $a = \min(k, x)$, $b = \max(k, x)$. The pair (a, b) determines k, so the operator Schmidt decomposition of $U(t)|_{\text{module}}$ is nothing but a re-indexing of the coefficient matrix A^t: rows (a, a'), columns (b, b'). The four index sectors land as follows.

index sector	row	column	content
$k, k' \le x$	(k, k')	(x, x)	the whole LL block, in one column
$k, k' > x$	(x, x)	(k, k')	the whole RR block, in one row
$k \le x < k'$	(k, x)	(x, k')	the crossing block C_{LR}
$k' \le x < k$	(x, k')	(k, x)	the crossing block C_{RL}

The two crossing blocks occupy disjoint rows and disjoint columns, and the remaining content is confined to the single row (x, x) and the single column (x, x). Hence, with no approximation,

$$\operatorname{rank} C_{LR} + \operatorname{rank} C_{RL} \;\le\; \chi_x(t) \;\le\; 2 + \operatorname{rank} C_{LR} + \operatorname{rank} C_{RL}. \tag{2}$$

The whole problem is now the rank of one crossing block.

3.2 Step 2: a last-passage identity, and the transfer rank

Proposition 2 (Transfer rank)**.** *Let A act on a finite-dimensional space and let $P_L + P_R = \mathbb{1}$ be complementary projectors. Then for every $t \geq 1$*

$$P_R A^t P_L \;=\; \sum_{s=0}^{t-1} \big(P_R A P_R\big)^{t-1-s} \big(P_R A P_L\big) \big(P_L A^s P_L\big), \tag{3}$$

and consequently

$$\operatorname{rank}\big(P_R A^t P_L\big) \;\leq\; t \cdot \operatorname{rank}\big(P_R A P_L\big).$$

Proof. Expand A^t as a sum over sequences of L/R labels, one per time step. Every sequence contributing to $P_R A^t P_L$ starts in L and ends in R, so it has a unique *last* time s at which the label is L, with $0 \leq s \leq t-1$. Grouping the sequences by that s gives exactly (3): the factor $P_L A^s P_L$ collects all histories up to time s (unrestricted in between), the factor $P_R A P_L$ is the step out of L at time $s+1$, and $(P_R A P_R)^{t-1-s}$ the remaining steps, which stay in R by construction. Each summand contains $P_R A P_L$ as a factor, so each has rank at most $\operatorname{rank}(P_R A P_L)$, and rank is subadditive. □

The identity is checked numerically to $\leq 2.2 \times 10^{-16}$ for the kink propagator at $N = 24, 30$ and $t \leq 10$, and for a random 10×10 unitary; it is elementary, but it is the step that converts a *static*, one-period quantity into a linear-in-t bound.

It remains to compute $\operatorname{rank}(P_R A P_L)$ for the kink walk. Group the kink positions into cells $\{2j-1, 2j\}$ and pass to

$$|p_j\rangle = \tfrac{1}{\sqrt{2}}\big(|2j-1\rangle + |2j\rangle\big), \qquad |m_j\rangle = \tfrac{1}{\sqrt{2}}\big(|2j-1\rangle - |2j\rangle\big).$$

In this basis the exact one-period propagator of W_{156} reads, in the bulk,

$$\begin{aligned} A\,|p_j\rangle &= \tfrac{1}{2}\big(|m_{j-1}\rangle + |m_j\rangle - |p_j\rangle + |p_{j+1}\rangle\big), \\ A\,|m_j\rangle &= \tfrac{1}{2}\big(|m_{j-1}\rangle - |m_j\rangle - |p_j\rangle - |p_{j+1}\rangle\big), \end{aligned} \tag{4}$$

i.e. a two-band quantum walk hopping by at most one cell per period. Its inter-cell transfer operators are

$$T_+ \;=\; \tfrac{1}{2}\,|p\rangle\big(\langle p| - \langle m|\big), \qquad T_- \;=\; \tfrac{1}{2}\,|m\rangle\big(\langle p| + \langle m|\big), \tag{5}$$

both of rank one (W_{198} is the mirror of W_{156} and reflection sends $k \mapsto l+1-k$, so its cell tiling is the shifted pairing $\{2j, 2j+1\}$ and the roles of p and m are exchanged). The coin state p is the sole right-moving channel and m the sole left-moving one: the walk is chiral, with one channel per chirality per cell. Measured directly, $\operatorname{rank}(P_R A P_L) = 1$ at *every* cut $x = 1, \ldots, l-1$, for both rules at $N = 12, 20, 24$, so Proposition 2 gives $\operatorname{rank} C_{LR} \leq t$, likewise for C_{RL}.

3.3 Step 3: the matching lower bound

Because $P_R A P_L = ab^{\mathsf{T}}$ is rank one, (3) is an explicit rank-t factorisation:

$$P_R A^t P_L \;=\; \sum_{s=1}^{t} u_s v_s^{\mathsf{T}}, \qquad u_s = (P_R A P_R)^{t-s} a, \qquad v_s = \big(P_L A^{s-1} P_L\big)^{\mathsf{T}} b.$$

If $\{u_s\}_{s \leq t}$ and $\{v_s\}_{s \leq t}$ are each linearly independent, the factorisation is UV^{T} with both factors of full column rank t, so the rank is exactly t. Independence follows from the light cone: by (5) the rightward amplitude per period is exactly $\frac{1}{2}$ and never vanishes, so u_s reaches exactly $t-s$ cells to the right of the cut with leading amplitude $2^{-(t-s)}$, and v_s exactly $s-1$ cells to the left. The reaches are pairwise distinct, hence each family is independent. Measured: $\dim \operatorname{span}\{u_s\} = \dim \operatorname{span}\{v_s\} = \operatorname{rank} = t$ for all $t \leq 9$ at $N = 24, 30$, both rules, with the factorisation reproducing $P_R A^t P_L$ to $\leq 2.2 \times 10^{-16}$.

3.4 The law

Combining (2) with $\operatorname{rank} C_{LR} = \operatorname{rank} C_{RL} = t$:

Theorem 1 (Kink module)**.** *For W_{156} and W_{198} on their maximal one-wall module of dimension l, and any cut x with $1 \le t < \min(x, l-x)$ — i.e. as long as the light cone has not reached either end of the chain —*

$$2t \;\le\; \chi_x(t) \;\le\; 2t+2, \qquad \textit{so} \qquad \chi_x(t) = \Theta(t).$$

The measured value is the middle one, and the saturation is a clean truncation:

$$\chi_x(t) \;=\; \min\left(2t+1,\; \chi_x^{\text{sat}}\right), \qquad t \ge 1, \tag{6}$$

with χ_x^{sat} a t-independent, cut-dependent ceiling. Verified exhaustively for both rules at $N = 12, 16, 20$, *every* cut $x = 1, \dots, N-1$ and t up to $3N$, and at $N = 24, 30$ for $x = 3,\ N/2,\ N/2+1$. At $t = 0$ the value is 2: the identity on the module is not a product operator across the cut, being $\mathbb{1}_{\text{left}} \otimes P_R + P_L \otimes \mathbb{1}_{\text{right}}$. This is the statement rev. 1 left open. The ceiling χ_x^{sat} is a finite-chain edge effect, small when the cut is near an end; no closed form for it is claimed here.

3.5 The full chain

On the full 2^N space the exact operator Schmidt rank of U^t across the middle bond is

$$\boxed{\chi(t) \;=\; 6t-7, \qquad 4 \le t \le \min(N/2,\, 12).} \tag{7}$$

The upper limit 12 is invisible at the sizes in this section — for $N \le 16$ it is $N/2$ that binds — and was found only at $N \ge 31$; see §6. Measured series (identical for W_{156} and W_{198}, obc0, $x = N/2$):

N	$\chi(t)$, $t = 0, 1, 2, \dots$	law holds for	first deviation
8	$1, 4, 8, 12, 17, 20, 21, \dots$	$t \le 4$	$t = 5$
10	$1, 4, 8, 12, 17, 23, 27, 30, 31, \dots$	$t \le 5$	$t = 6$
12	$1, 4, 8, 12, 17, 23, 29, 33, 37, 41, \dots$	$t \le 6$	$t = 7$
14	$1, 4, 8, 12, 17, 23, 29, 35, 39, 43$	$t \le 7$	$t = 8$
16	$1, 4, 8, 12, 17, 23, 29, 35, 41, 45, 49$	$t \le 8$	$t = 9$

The increments are $3, 4, 4, 5$ and then a constant 6; the deviations past $t = N/2$ are the finite-chain saturation, and the window grows by one period per two sites, exactly as a light cone should. The $N = 14$ and $N = 16$ rows were computed with a sparse propagator and a randomised rank estimator (oversampled to 160 probe vectors, so the reported rank is exact whenever it is below the cap, which it always is).

Remark 4 (The slope 6 is not yet derived — and three routes are closed)*.* Theorem 1 explains the linearity and gives slope 2 per straddling gap. The factor 3 relating that to (7) is not derived here, and the obvious derivations do not work:

1. *Not a sum over Krylov sectors.* The reshuffle is supported on entry (y, y') only when y, y' share a sector, so it splits into blocks of disjoint support; but the left Schmidt vectors are shared *between* sectors, and the sum is wildly larger than the total. At $N = 12$ (377 sectors) $\sum_S \chi_S(t) = 473, 569, 700, 747, 763, 749, 731, \dots$ against $\chi(t) = 1, 4, 8, 12, 17, 23, 29, \dots$ — not even monotone. Fragmentation alone does not make the MPO small; the sharing does.

2. *Not a sum over growing sectors either.* Only 12 of the 377 sectors have a χ_S still growing at $t = 6$, each with slope 2 while unsaturated, for a total of 24 where the full chain increments by 6.

3. *Not an accumulating channel basis.* The left Schmidt subspace does not grow as a filtration: projecting the $t{+}1$ subspace against the t one leaves 20–21 new directions out of 23–29, so the subspace rotates almost completely each period and there is no fixed set of "six new channels" to count. Every Schmidt vector also acts non-trivially out to the far end of the half-chain, so the new directions cannot be localised at the light-cone edge.

The slope is therefore a genuinely global count over the wall grammar, and deriving it needs the MPO of U^t itself rather than any sector-wise argument.

4 The ceiling

$\chi(t)$ does not grow without bound. It saturates, and the saturation value is closed-form. This is the sharper statement: a growth law describes a window and invites the objection that the window is merely the light cone before saturation, whereas a ceiling *polynomial in N and uniform in t* says the propagator is efficiently representable at all times.

4.1 Exact ranks, with no threshold at all

Every amplitude in this project lies in $\mathbb{Z}[1/\sqrt{2}]$: it is $(a + b\sqrt{2})/2^m$ with $a, b \in \mathbb{Z}$. So the rank of the reshuffle is an algebraic quantity and need never be read off a singular-value spectrum. Choose a prime $p \equiv \pm 1 \pmod 8$, so that 2 is a quadratic residue and $\sqrt{2}$ exists in $\mathbb{F}_p$, and map the whole computation — U, its powers, the reshuffle, the elimination — into $\mathbb{F}_p$. Keeping $2^N(p-1)^2 < 2^{53}$ makes float64 matrix multiplication *exact*, so the powers still run at BLAS speed. Since $\mathrm{rank}_{\mathbb{F}_p} \le \mathrm{rank}_{\mathbb{Q}(\sqrt{2})}$ with equality off a finite set of p, agreement across two primes is the usual certificate.

rule	closed form	$N{=}7$	$N{=}9$	$N{=}11$	$N{=}13$
W_{156}	$(N^2+15)/4$	16	24	34	46
W_{198}	$(N^2+15)/4$	16	24	34	46
W_{108}	$F(k)F(k{+}1)$	15	40	104	273
W_{201}	$F(k)\big(F(k){+}3F(k{-}1)\big)$	27	70	184	481

Table 4: Plateau of χ, maximised over bonds, computed exactly over $\mathbb{Q}(\sqrt{2})$ via $\mathbb{F}_p$; $k = (N{+}1)/2$. All sixteen entries agree with the closed form and agree at both primes (1048559/1048447, and 741431/741409 at $N{=}13$). Data: `analytics/r27_exact_ranks.json`.

The contrast that matters is between the two gate slots and is **polynomial versus exponential in** N: $(N^2+15)/4$ against $\sim \varphi^N$. That maps onto the defect picture of Proposition 1 — $p = 1$ kink module against extensive p — with the *ceiling*, not the slope, as the diagnostic. The closed forms are due to the MPO measurement on the experimental side of the project, fitted there; what is contributed here is their exact confirmation.

4.2 The maximum is off-centre

W_{156} is not reflection symmetric — its mirror is W_{198} — so at odd N the two candidate middle bonds need not agree, and they do not. The per-bond profile of the saturated χ at $N = 9$ is

$$2,\ 6,\ 14,\ 22,\ \mathbf{24},\ 17,\ 9,\ 4 \qquad (x = 1, \ldots, 8),$$

with the maximum at $x = (N{+}1)/2$ for W_{156} and at $x = (N{-}1)/2$ for W_{198}; the two central bonds differ by exactly 2, so the middle cut gives $(N^2{+}15)/4 - 2$. The closed form is therefore strictly a *maximum over bonds.* The two rules' profiles are mirror images of one another, never equal.

4.3 The gate sets the dynamics, not the ceiling

Under $V = X$ the circuit is a permutation of basis states, so $U(t)$ is a permutation matrix, the reshuffle is a 0/1 matrix, and its rank is an exact integer — there is no threshold to choose even in principle. Computed that way, the largest χ over $t \leq 400$ is

$$24,\ 34,\ 46,\ 60 \qquad \text{at } N = 9, 11, 13, 15,$$

for both X_{156} and X_{198}: *identical to the Hadamard ceiling.* Replacing the Hadamard by a classical gate changes the dynamics completely ($U(t)$ becomes periodic and χ never locks) but leaves $(N^2{+}15)/4$ untouched, which places the ceiling in the wall grammar rather than in the gate.

Remark 5 (A supremum only where a period was swept). At these sizes the permutation did not close its period within 400 steps, so the numbers above are maxima over a sampled window. The experimental side has since swept a full period at $N = 9$ (ord $= \mathrm{lcm}(1, \ldots, 9) = 2520$, halved by $\chi(t) = \chi(\mathrm{ord} - t)$, which holds because $U(\mathrm{ord} - t) = U(t)^{\mathsf{T}}$ and transposition preserves the reshuffle rank) and obtains 24 as a true supremum. For $N \geq 11$ neither side has swept a period, and "$\sup_t$" should be read as a windowed maximum.

5 Reflection: where the slope comes from

Rev. 2 recorded the slope 6 as measured and excluded three derivations of it. A mechanism has since been proposed — *between two stationary walls a moving domain wall propagates, reflects, and returns* — and it survives a direct test.

Krylov sectors are conserved, so restricting to a union of sectors is legitimate and the Schmidt rank of $U(t)P$ is well defined. Bucket the sectors of W_{156} by the length L of the gap straddling the cut, with walls identified as the sites frozen across the sector. All ranks below are exact mod p.

bucket	$\chi(t)$, $t = 1, 2, \ldots$	reading
$L = 0$ (cut in a wall)	$4, 4, 4, 4, \ldots$ to t=40	walls contribute a purely t-independent constant
$L = 11$ (one long gap)	increments $2, 2, 2, 2$	the module law, Theorem 1, recovered inside the full chain
$L = 3$	$6, 8, 8, 8, 6, 4,\ 6, 8, 8, 8, 6, 4, \ldots$	**non-monotone, exactly periodic with period** 6, verified to t=40

The third line is reflection observed directly: χ falls and returns, because the kink bounces off both walls and comes back to a coherent configuration. Every other bucket saturates to a constant; $L = 3$ is the only one that never does.

5.1 The ladder

Admitting only gaps with $L \geq L_{\min}$, the early-time slope depends on the number k of admitted length classes and not on N:

k classes	1	2	3	4	5	6
slope, $N = 12$	2	2	4	6	—	—
slope, $N = 14$	2	2	4	6	8	10

$$\text{slope} \;=\; 2(k-1) \quad (k \geq 2), \qquad 2 \quad (k = 1). \tag{8}$$

Both sizes agree at every k they share. Each additional gap length contributes exactly one module's worth, 2. The full chain's 6 corresponds to four effective classes.

Why the full chain does not simply give $2(k_{\max} - 1)$ is the same mechanism: short gaps contribute steeply and then *stop*, because their kink has bounced and explored the whole gap, while long gaps keep contributing. The observed 6 is the residual after the short gaps have gone quiet and before the long ones do. Reflection sets both the contribution and its cutoff time, which is why the slope is a finite N-independent constant instead of growing with the number of available gap lengths.

Remark 6 (Measured, not counted — in both gates)*.* (8) is a measured law, not a derivation. The buckets share Schmidt vectors heavily, so "four classes × two" describes the ladder without counting it. Under the Hadamard at $N = 12$, $t = 5$ the per-bucket ranks sum to 161 against a true χ of 23. The same holds under $V = X$, where the ranks are exact integers and the temptation to treat the decomposition as sound is strongest — measured on the experimental side at $N = 11$:

t	1	2	3	4	5	6	7	8
$\sum_L \chi_L$	64	88	99	100	95	88	76	71
χ	4	8	12	17	23	26	26	26
ratio	16	11	8.2	5.9	4.1	3.4	2.9	2.7

Exactness of the ranks says nothing about additivity of the decomposition. This is the same sharing that defeated the three derivations excluded in rev. 2, and it remains the obstacle in both gates.

5.2 The two gates behave differently, and only one has a period law

Under $V = X$ the same bucketing has been carried out on the experimental side, where the ranks are exact integers for free. There *every* bucket is a triangular wave — rise at slope 2, plateau, symmetric fall — with period exactly $L + 1$, verified for $L = 2, \ldots, 9$ at $N = 11, 13$ and both rules, with the short-gap curves independent of N. That derives two facts previously only measured: the period of $\chi(t)$ divides $\mathrm{lcm}(3, \ldots, N) = \mathrm{lcm}(1, \ldots, N)$, which is the observed order of the permutation; and the ceiling is reached only when every bucket peaks together, which is why it is visited on a few percent of steps and why χ falls back to exactly 4 — that 4 is literally the $L = 0$ wall constant showing through. Both of those are heuristic readings rather than derivations, for the non-additivity reason recorded above; what genuinely follows is that a permutation has finite order, so $\chi(t)$ must be periodic and cannot converge, and $\mathrm{lcm}(3, \ldots, N) = \mathrm{lcm}(1, \ldots, N)$ is a consistency check on the observed order rather than a proof of it.

The Hadamard has no such law. Repeating the measurement here, exactly mod p, for both rules at $N = 12$ and $N = 14$, out to $t = 80$:

L	period	behaviour
0	—	constant at 4
3	$\mathbf{6} = 2L$	the only oscillating bucket, at either size
all other L	—	saturate to a constant

The $L = 3$ curve is $6, 8, 8, 8, 6, 4$ repeating, *byte-identical* at $N = 12$ and $N = 14$: the short-gap component is N-independent here exactly as it is under $V = X$. What differs is only that under $V = X$ every L behaves this way, whereas here $L = 3$ is alone.
So $L + 1$ is a statement about permutation circuits, not about the Hadamard, and even the period of the one oscillating bucket differs ($6 = 2L$ here against $L + 1 = 4$ there). Under the Hadamard the amplitudes spread and the rank fills its bucket instead of returning; only $L = 3$ is small enough to keep coming back. What survives across both gates is the part that matters for the mechanism: walls give a t-independent constant, the longest gap rises at slope exactly 2, and *at least one* short-gap component never saturates. The last clause must be stated in that weak form: under the Hadamard $L = 2$ — the shortest gap there is — saturates like all the others, and $L = 3$ alone oscillates. "Short gaps do not saturate" would be false here, and it is the plural version that would make the prediction below sound overdetermined.

Remark 7 (A prediction, and its honest weight). $L = 3$ is the only Hadamard bucket that never saturates. If it leaves a residual periodic component riding on the linear law, the departure from (7) should *repeat* rather than grow: the excess at $t = 19, 20, 21$ should echo $t = 13, 14, 15$. Two caveats. It is testable only at $N \geq 31$, beyond the reach of the 2^N methods used here, and beyond the reach of double precision at $t \geq 16$, so it needs exact elimination in place of SVD compression. And the prediction rests on a *single* bucket, not on a family — under $V = X$ a periodic correction would be overdetermined, but under the Hadamard everything else has gone quiet by then. It is a named test, not a strong expectation.

Remark 8 (A control). The same bucketing on W_{108} shows no such structure: the buckets are non-periodic and irregular, reaching rank 63 where W_{156} stays under 18. Single-kink reflection is therefore specific to the domain-wall slot, as it must be if that slot is the $p = 1$ case and the symmetric slot has extensive defect number. A mechanism that also "explained" W_{108} would be suspect.

6 Method: ranks without a threshold

6.1 Why the window is $\min(N/2, 12)$

Two departures from (7) exist and are cleanly separable. The first is a *downward*, finite-size roll-off at $t \sim (N+1)/2$ — the light cone reaching a chain end — which is what the sizes in §3 see. The second is an *upward*, N-independent departure beginning at $t = 13$, reported by the MPO measurement at $N \geq 31$: the excess over $6t - 7$ is $+1$ at t=13 and $+2$ at t=14 (both certified by a singular-value gap of 7.3×10^3 and 6.9×10^2), $+3$ at t=15 marginal, and *retracted* for $t \geq 16$ where the gap collapses to order unity and the rank is not defined in double precision. Below $N \approx 31$ the two effects cancel. Nothing in this report reaches $N \geq 31$, so the second departure is recorded as that measurement's, not as verified here.

6.2 The trap, three times

Three independent pipelines each manufactured a feature by holding a singular-value cutoff *fixed* while the spectrum moved underneath it, and they failed in both directions:

pipeline	spectrum	artefact
kink module, cutoff 10^{-9}	genuine tail decays $\sim 5.6\times$ per period	a false departure from $t = 14$
Hadamard MPO, cutoff 10^{-10}	same, $\sim 8\times$ per period	a false dip at $t = 12$
X-gate MPO, cutoff 10^{-14}	genuine tail *flat* at 10^{-4}	57 phantom channels (161 vs 104)

The module case is worth stating in full because it nearly entered this report as a result. At $N = 60$, cut $x = 30$, a float64 read at 10^{-9} reports $\operatorname{rank} C_{LR} < t$ from $t = 14$ on, N-independently ($N = 60, 80$ identical; $N = 200, 400$ identical) — which would have looked like striking confirmation of the $t = 13$ departure in a second sector from a second code path. It is false. The exact rank over $\mathbb{Q}(\sqrt{2})$ is t for every $t = 9, \ldots, 20$, and the spectrum shows why:

t	σ_t/σ_1	σ_{t+1}/σ_1	gap	rank at 10^{-9}
12	1.8×10^{-8}	1.2×10^{-16}	1.6×10^{8}	correct
13	3.3×10^{-9}	1.3×10^{-16}	2.5×10^{7}	correct
14	5.9×10^{-10}	1.5×10^{-16}	3.8×10^{6}	short by 1
17	3.3×10^{-12}	1.6×10^{-16}	2.1×10^{4}	short by 2
20	1.8×10^{-14}	1.5×10^{-16}	1.2×10^{2}	short by 3

The gap to the true null space never closes below 10^2: the rank was never ambiguous, the threshold was simply in the wrong place.

Remark 9 (Two rules). **(i)** The certificate is σ_r/σ_{r+1} at the claimed rank, never the value of a cutoff and never a sweep of it — a sweep of the *plateau* says nothing about the *departure*, which is where the doubt lives. **(ii)** Where the object is integral or lies in a number ring, take the algebraic rank and the question does not arise. That is free for $V = X$, where the reshuffle is 0/1, and available for the Hadamard case at any size where the matrix can be formed, by the $\mathbb{F}_p$ route of §4.

Remark 10 (Which rank is the physics). Because the genuine tail decays geometrically, "the rank of $U(t)$" is threshold-dependent and one must say which is meant. (7) and (6) are *algebraic* rank, the right choice for a law. The quantity governing simulability is the ϵ-rank at a practical ϵ, and it is strictly *smaller*: on the module at $\epsilon = 10^{-9}$ it stalls near 17 while the algebraic rank climbs. Efficient representability is therefore a stronger claim than the algebraic law, not a weaker one.

7 What would settle it

1. **A counting proof of the slope.** (8) gives the mechanism and the arithmetic, but the buckets share Schmidt vectors, so the ladder is measured rather than derived. Turning $2(k-1)$ into a theorem needs the MPO transfer structure of U^t at the cut.

2. **Is the departure periodic?** The excess over $6t-7$ at $t = 19, 20, 21$ against $t = 13, 14, 15$ decides whether the $t = 13$ effect is short-gap reflection or a genuine onset. It needs $N \geq 31$ and exact ranks at $t \geq 16$, i.e. rank-revealing elimination over $\mathbb{F}_p$ in place of SVD compression.

3. **Beyond one straddling gap.** Theorem 1 is the $p = 1$ case; confirming $O(t^p)$ on a module with $p = 2$ would turn defect counting from an explanation of two data points into a law.

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